ExamGOAL
PYQ PDF

Free · With answer key

GATE ECE · DIGITAL CIRCUITS · CHAPTER-WISE PYQS

Sequential Circuits

GATE ECE Digital Circuits · Previous Year Questions with Answer Key

79	41	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	32 questions
Marks 2	37 questions
Marks 5	6 questions
Marks 8	4 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve** — Print it, or work through it on screen.
2. **Check** — The answer key is on the last pages.
3. **Understand** — Scan the QR code beside any question for its full solution.

Keep practising on the ExamGOAL app

Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.

GET IT ON Google Play

Marks 1 32 questions

Q1. GATE ECE 2026

A binary ripple counter is designed to count $(0)_{10}$ to $(64)_{10}$. Which of the following is/are the number of flip-flops required to design the counter?

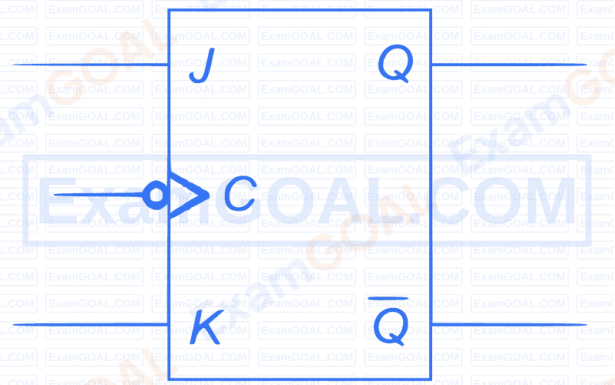
(A) 6 (B) 7 (C) 4 (D) 5

Open solution

Q2. GATE ECE 2026

The negative edge triggered JK flip-flop in the Figure has J and K inputs tied to Logic High and a square wave of 10 cycles/second is applied to its clock (C) input. The frequency of the output Q (in cycles/second) is ____ .

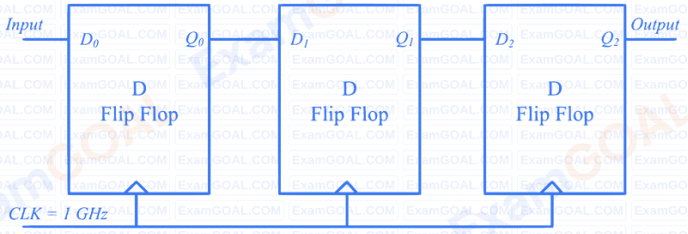
(rounded off to two decimal places)



Open solution

Q3. GATE ECE 2023

The synchronous sequential circuit shown below works at a clock frequency of 1 GHz. The throughput, in Mbits/s, and the latency, in ns, respectively, are



(A) 1000, 3 (B) 333.33, 1 (C) 2000, 3 (D) 333.33, 3

Open solution

Q4. GATE ECE 2018

A traffic signal cycles from GREEN to YELLOW, YELLOW to RED and RED to GREEN. In each cycle, GREEN is turned on for 70 seconds, YELLOW is turned on for 5 seconds and the RED is turned on for 75 seconds. This traffic light has to be implemented using a finite state machine (FSM). The only input to this FSM is a clock of 5 second period. The minimum number of flip-flops required to implement this FSM is _____.

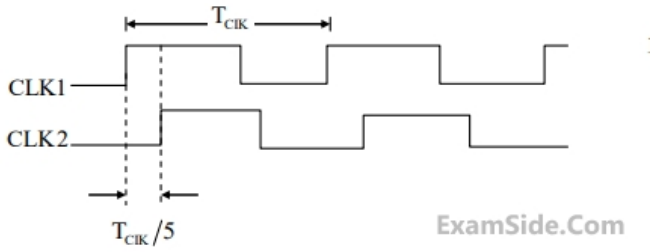
Open solution

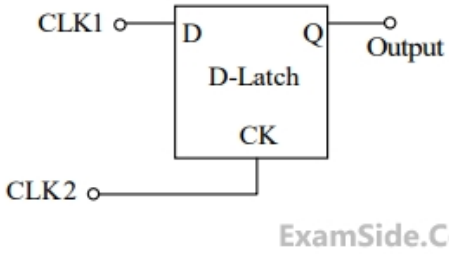
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q5. GATE ECE 2017 Set 1

Open solution

Consider the D-Latch shown in the figure, which is transparent when its clock input **CK** is high and has zero propagation delay. In the figure, the clock signal **CLK1** has a 50% duty cycle and **CLK2** is a one-fifth period delayed version of **CLK1**. The duty cycle at the output latch in percentage is _____.

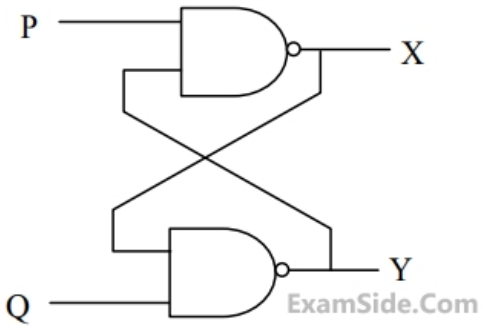




Q6. GATE ECE 2017 Set 1

Open solution

In the latch circuit shown, the NAND gates have non-zero, but unequal propagation delays. The present input condition is: P = Q = "0". If the input condition is changed simultaneously to P = Q = "1", the outputs X and Y are



(A) X = '1', Y = '1'

(B) either X = '1', Y = '0' or X = '0', Y = '1'

(C) either X = '1', Y = '1' or X = '0', Y = '0'

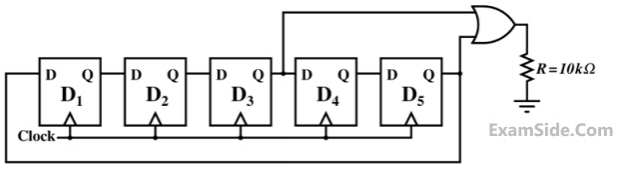
(D) X = '0', Y = '0'

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q7. GATE ECE 2016 Set 2

Open solution

Assume that all the digital gates in the circuit shown in the figure are ideal, the resistor R = 10 $k\Omega$ and the supply voltage is 5 V. The D flip-flops D_1, D_2, D_3, D_4 and D_5 are initialized with logic values 0, 1, 0,1 and 0, respectively. The clock has a 30% duty cycle.

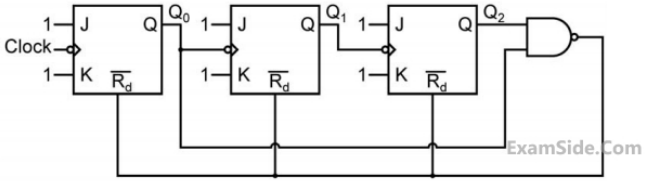


The average power dissipated (in mW) in resistor R is _____.

Q8. GATE ECE 2015 Set 3

Open solution

The circuit shown consists of J-K flip-flops, each with an active low asynchronous reset ($\overline{R_d}$ input). The counter corresponding to this circuit is



(A) a modulo-5 binary up counter

(B) a modulo-6 binary down counter

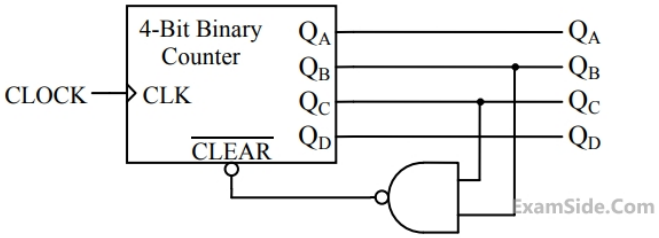
(C) a modulo-5 binary down counter

(D) a modulo-6 binary up counter

Q9. GATE ECE 2015 Set 2

Open solution

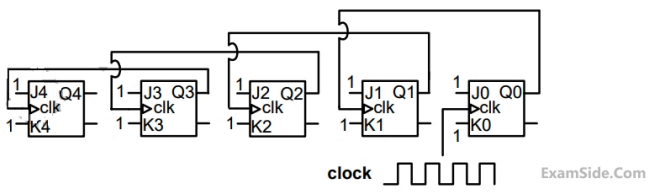
A mod-n counter using a synchronous binary up-counter with synchronous clear input is shown in the figure. The value of n is _____.



Q10. GATE ECE 2014 Set 1

Open solution

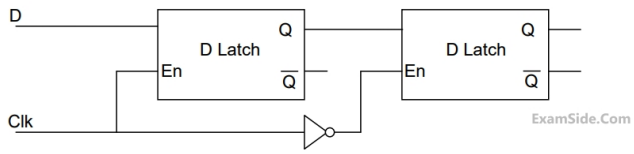
Five JK flip - flops are cascaded to form circuit shown in figure. Clock pulses at a frequency of 1 MHz are applied as shown. The frequency (in kHz) of the waveform at Q_3 is _____.



Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q11. GATE ECE 2014 Set 3

The circuit shown in the figure is a



Open solution

(A) Toggle Flip Flop

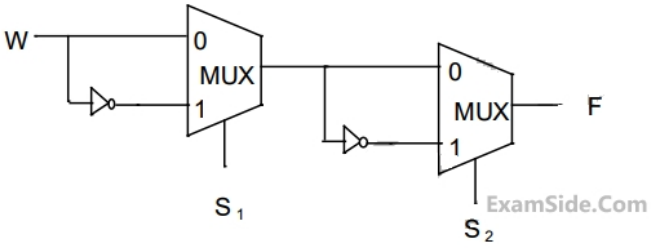
(B) JK Flip Flop

(C) SR Latch

(D) Master-Slave D Flip Flop

Q12. GATE ECE 2014 Set 3

Consider the multiplexer based logic circuit shown in the figure.



Open solution

Which one of the following Boolean functions is realized by the circuit?

(A) F = $W\overline{S_1 S_2}$

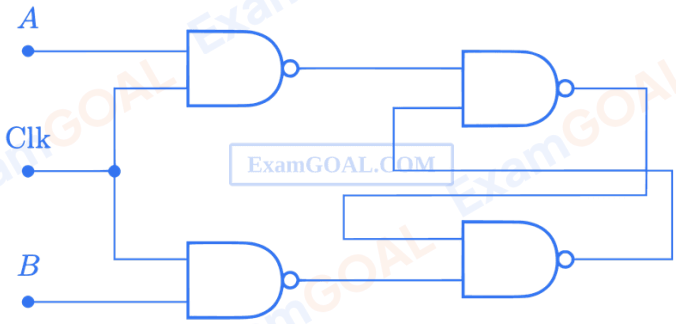
(B) F = $WS_1 + WS_2 + S_1S_2$

(C) F = $\overline{W} + S_1 + S_2$

(D) F = $W \oplus S_1 \oplus S_2$

Q13. GATE ECE 2012

Consider the given circuit. In this circuit, the race around



Open solution

(A) does not occur

(B) occurs when CLK = 0

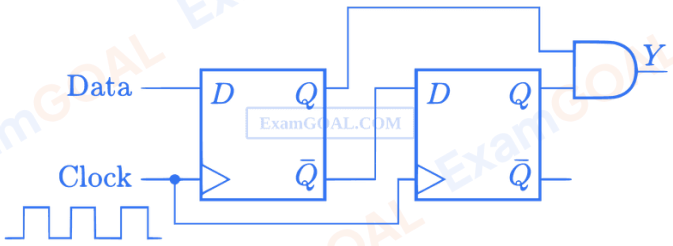
(C) occurs when CLK = 1 and A = B = 1

(D) occurs when CLK = 1 and A = B = 0

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q14. GATE ECE 2011

When the output Y in the circuit below is '1', it implies that data has

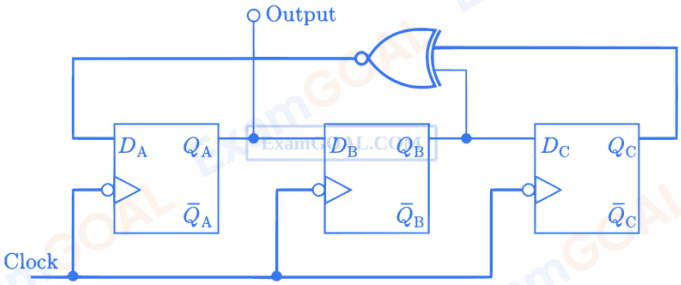


Open solution

(A) changed from 0 to 1
(B) changed from 1 to 0
(C) changed in either direction
(D) not changed

Q15. GATE ECE 2010

Assuming that all flip-flops are in reset condition initially, the count sequence observed at Q_A in the circuit shown is



Open solution

(A) 0010111...
(B) 0001011...
(C) 0101111...
(D) 0110100...

Q16. GATE ECE 2005

The present output Q_n of an edge triggered JK flip-flop is logic 0. If J=1, then Q_{n+1}

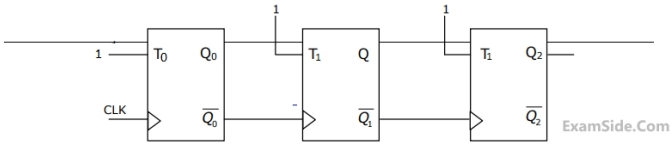
Open solution

(A) can not be determined
(B) will be logic 0
(C) will be logic 1
(D) will race around

Q17. GATE ECE 2005

The given figure shows a ripple counter using positive edge triggered flip-flops. If the present state of counter is Q_2 Q_1 Q_0 = 011, then its next state (Q_2 Q_1 Q_0) will be

Open solution



(A) 010
(B) 100
(C) 111
(D) 101

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q18. GATE ECE 2004

A master slave flip-flop has the characteristic that

(A) change in the input immediately reflected in the output

(B) change in the output occurs when the state of the master is affected

(C) change in the output occurs when the state of the slave is affected

(D) both the master and the slave states are affected at the same time

Open solution

Q19. GATE ECE 2004

Choose the correct one from among the alternatives A, B, C, D after matching an item from Group 1 with the most appropriate item in Group 2.

Group1

P. shift register

Q. Counter

R. Decoder

Group2

1. Frequency division
2. Addressing in memory chips
3. Serial to parallel data conversion

(A) P-3, Q-2, R-1 (B) P-3, Q-1, R-2 (C) P-2, Q-1, R-3 (D) P-1, Q-3, R-2

Open solution

Q20. GATE ECE 2003

A 0 to 6 counter consist of 3 flip-flops and a combination circuit of 2 input gate(s). The combination circuit consists of

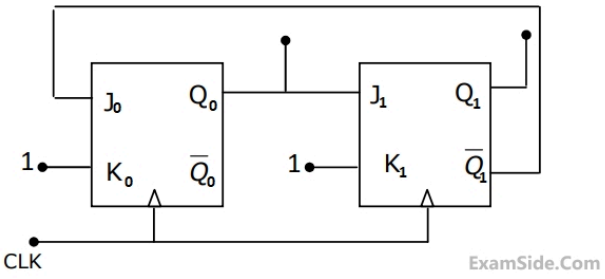
(A) one AND gate (B) one OR gate

(C) one AND gate and one OR gate (D) two AND gates

Open solution

Q21. GATE ECE 1998

Figure shows a mod-K counter, Here K is equal to



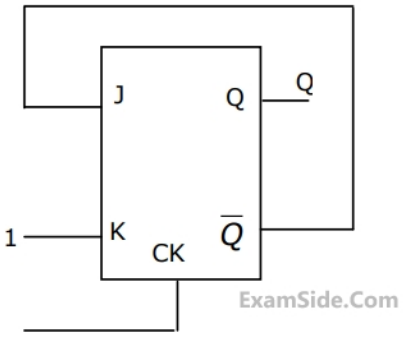
(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q22. GATE ECE 1997

In a J_K flip-flop, we have J=$\overline{Q}$ and K=1 (see figure). Assuming the flip-flop was intially cleared and then clocked for 6 pulses, the sequence at the Q output will be



(A) 010000 (B) 011001 (C) 010010 (D) 010101

Open solution

Q23. GATE ECE 1995

A switch-tail ring counter is made by using a single D flip-flop. The resulting circuit is a

(A) SR flip-flop (B) JK flip-flop (C) D flip-flop (D) T flip-flop

Open solution

Q24. GATE ECE 1995

An R-S latch is

(A) Combinational circuit (B) Synchronous sequential circuit.
(C) One bit memory element (D) One clock delay element.

Open solution

Q25. GATE ECE 1994

Synchronous counters are _____ than the ripple counters.

Open solution

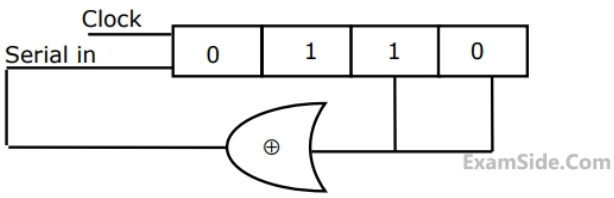
Q26. GATE ECE 1993

A pulse train with a frequency of 1 MHz is counted using a modulo-1024 ripple-counter built with J-K flip flops. For proper operation of the counter, the maximum permissible propagation delay per flip flop stage is _____ n sec.

Open solution

Q27. GATE ECE 1992

The initial contents of the 4-bit serial-in-parallel-out, right-shift, Shift Register shown in figure is 0110. After three clock pulses are applied, the contents of the Shift Register will be



(A) 0000 (B) 0101 (C) 1010 (D) 1111

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q28. GATE ECE 1991

A SR FLIP-FLOP can be converted into a T FLIP-FLOP by connecting ___ to Q and ___ to $\overline{Q}$.

Open solution

Q29. GATE ECE 1990

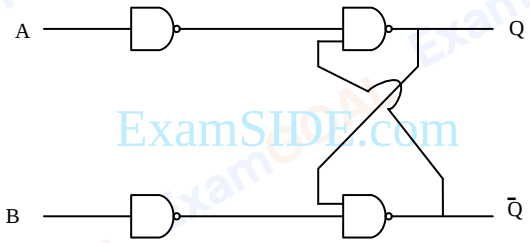
A 4 bit modulo-16 ripple counter uses JK flip-flops. If the propagation delay of each FF is 50ns, the maximum clock frequency that can be used is equal to:

(A) 20 MHz (B) 10 MHz (C) 5 MHz (D) 4 MHz

Open solution

Q30. GATE ECE 1988

The circuit given below is a



ExamSIDE.com

(A) J-K Flip-flop (B) Johnson's counter
(C) R-S latch (D) None of above

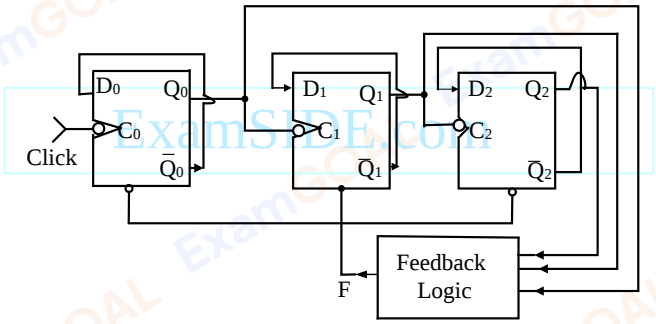
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q31. GATE ECE 1987

Open solution

A ripple counter using negative edge-triggered D-flip flops is shown in Fig.1. The flip-flops are cleared to '0' by a '0' at the R input. The feedback logic is to be designed to obtain the count sequence shown in the same figure. The correct feedback logic is:



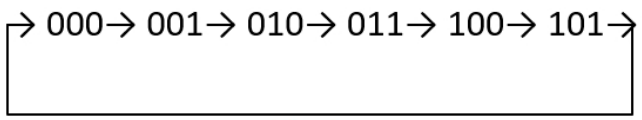


Fig:1

ExamSide.Com

(A) $F = \overline{Q_2 Q_1 \overline{Q_0}}$

(B) $F = Q_2 \overline{Q_1 Q_0}$

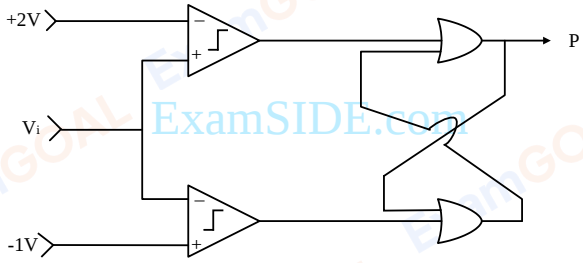
(C) $F = \overline{Q_2 Q_1} Q_0$

(D) $F = \overline{Q_2 Q_1 Q_0}$

Q32. GATE ECE 1987

Open solution

Choose the correct statements relating to the circuit of figure



(A) For V_i = -2V, P=0

(B) For V_i = +3V, P=0

(C) For V_i = 0V, P=0 always

(D) For V_i = 0V, P can be either 0 or 1.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

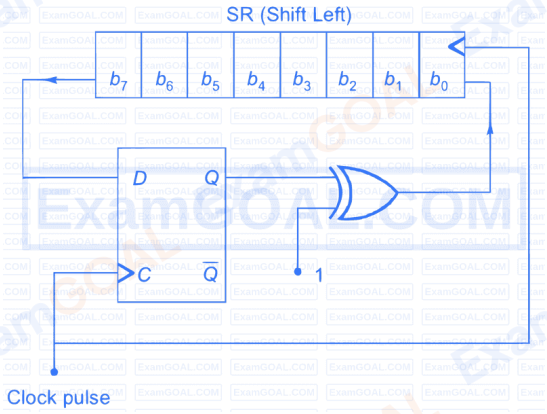
Marks 2 37 questions

Q33. GATE ECE 2026

A shift-left Shift Register (SR) and a D flip-flop are connected to a synchronized clock as shown in the Figure. Assume that the SR and D flip-flops are initially cleared and the XOR gate has no propagation delay.

Which of the following options gives the correct binary representation $(b_7b_6b_5b_4b_3b_2b_1b_0)$ of the content of the shift register immediately after the 5th clock transition (positive edge)?

Open solution



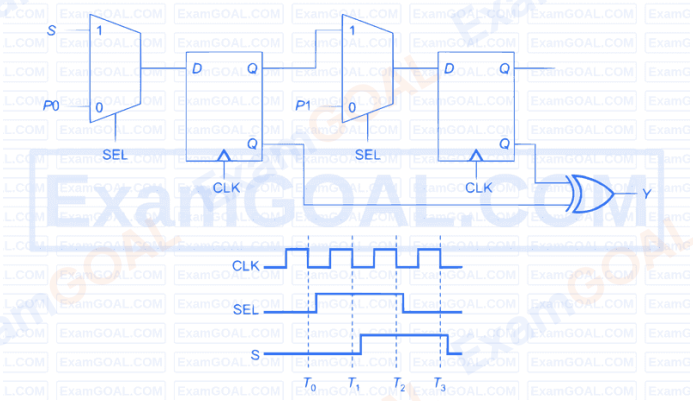
(A) 00011111 (B) 10111111 (C) 00111111 (D) 11000011

Q34. GATE ECE 2025

A positive-edge-triggered sequential circuit is shown below. There are no timing violations in the circuit. Input $P0$ is set to logic ' 0 ' and $P1$ is set to logic ' 1 ' at all times. The timing diagram of the inputs SEL and S are also shown below.

The sequence of output Y from time T_0 to T_3 is .

Open solution



(A) 1011 (B) 0100 (C) 0010 (D) 1101

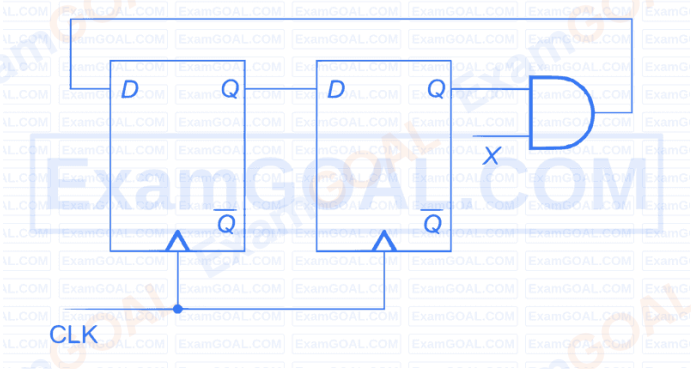
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q35. GATE ECE 2025

Open solution

In the circuit shown below, the AND gate has a propagation delay of 1 ns . The edgetriggered flip-flops have a set-up time of 2 ns , a hold-time of 0 ns , and a clock-to-Q delay of 2 ns .

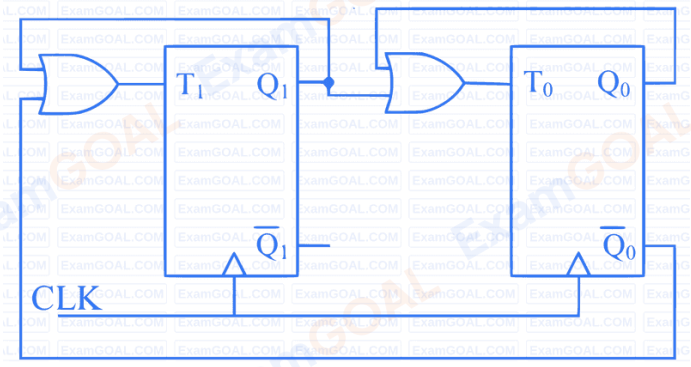
The maximum clock frequency (in MHz , rounded off to the nearest integer) such that there are no setup violations is_____ .



Q36. GATE ECE 2024

Open solution

The sequence of states (Q_1Q_0) of the given synchronous sequential circuit is _____.



(A) 00 → 10 → 11 → 00

(B) 11 → 00 → 10 → 01 → 00

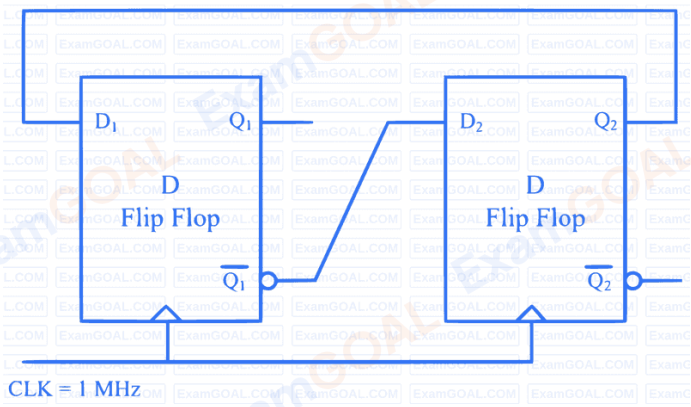
(C) 01 → 10 → 11 → 00 → 01

(D) 00 → 01 → 10 → 00

Q37. GATE ECE 2023

Open solution

In a given sequential circuit, initial states are $Q_1 = 1$ and $Q_2 = 0$. For a clock frequency of 1 MHz, the frequency of signal Q_2 in kHz, is _____ (rounded off to the nearest integer).

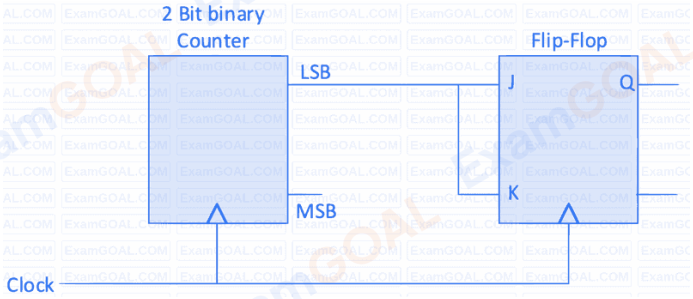


Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q38. GATE ECE 2022

Open solution

for the circuit shown, the clock frequency is f_0 and the duty cycle is 25%. For the signal at the Q output of the Flip-Flop, _____.



(A) frequency is $f_0/4$ and duty cycle is 50%

(B) frequency is $f_0/4$ and duty cycle is 25%

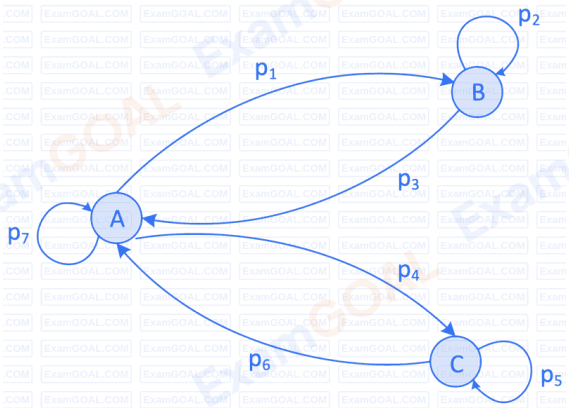
(C) frequency is $f_0/2$ and duty cycle is 50%

(D) frequency is f_0 and duty cycle is 25%

Q39. GATE ECE 2022

Open solution

A state transition diagram with states A, B, and C, and transition probabilities p_1, p_2, ..., p_7 is shown in the figure (e.g., p_1 denotes the probability of transition from state A to B). For this state diagram, select the statements which is/are universally true.



(A) $p_2 + p_3 = p_5 + p_6$

(B) $p_1 + p_3 = p_4 + p_6$

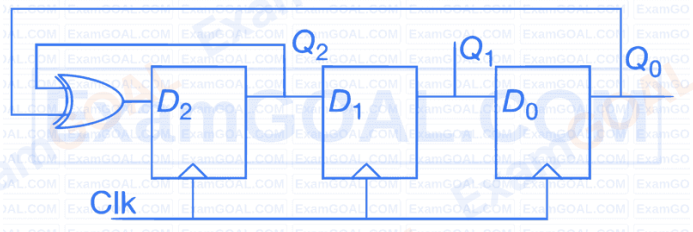
(C) $p_1 + p_4 + p_7 = 1$

(D) $p_2 + p_5 + p_7 = 1$

Q40. GATE ECE 2021

Open solution

The propagation delay of the exclusive-OR (XOR) gate in the circuit in the figure is 3 ns . The propagation delay of all the flip-flops is assumed to be Zero. The clock (Clk) frequency provided to the circuit is 500 MHz .



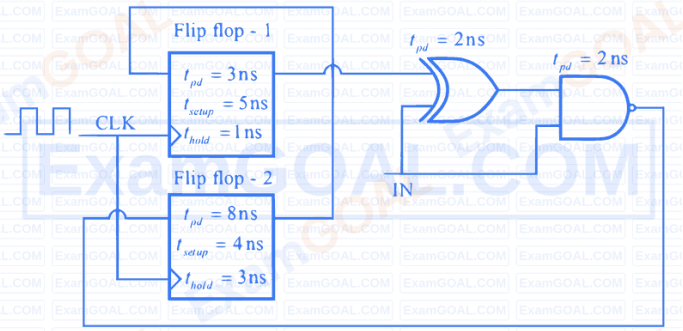
Starting from the initial value of the flip-flop outputs $Q_2Q_1Q_0 = 111$ with $D_2 = 1$, the minimum number of triggering clock edges after which the flip-flop outputs $Q_2Q_1Q_0$ becomes 100 (in integer) is ____

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q41. GATE ECE 2020

Open solution

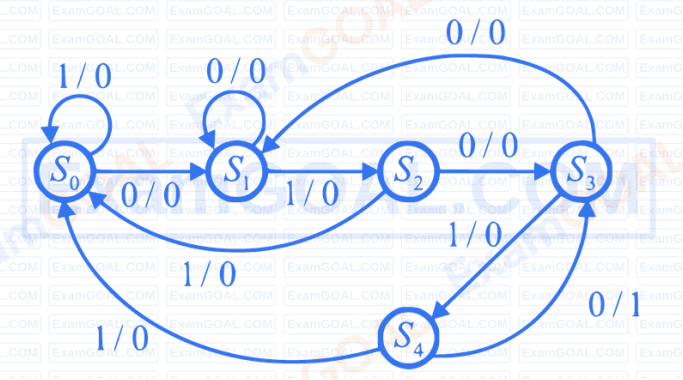
For the components in the sequential circuit shown below, t_{pd} is the propagation delay, t_{secup} is the setup time and t_{hold} is the hold time. The maximum clock frequency (rounded off to the nearest integer), at which the given circuit can operate reliably, is ____ MHz.



Q42. GATE ECE 2020

Open solution

The state diagram of a sequence detector is shown below. State S_0 is the initial state of the sequence detector. If the output is 1 , then



(A) the sequence 01110 is detected.

(B) the sequence 01001 is detected.

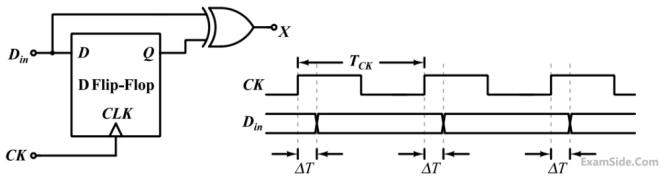
(C) the sequence 01011 is detected.

(D) the sequence 01010 is detected.

Q43. GATE ECE 2018

Open solution

In the circuit shown below, a positive edge-triggered D Flip-Flop is used for sampling input data D_{in} using clock CK. The XOR gate outputs 3.3 volts for logic HIGH and 0 volts for logic LOW levels. The data bit and clock periods are equal and the value of $\frac{\Delta T}{T_{CK}}$ = 0.15, where the parameters ΔT and T_{CK} are shown in the figure. Assume that the Flip-Flop and the XOR gate are ideal.



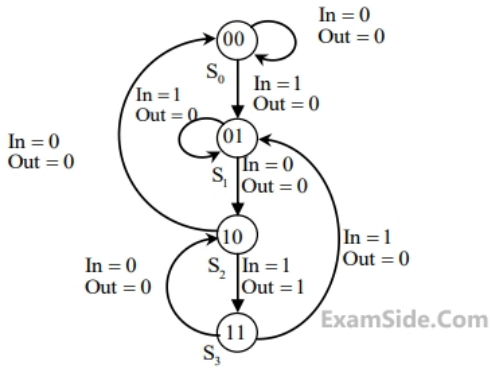
If the probability of input data bit (D_{in}) transition in each clock period is 0.3, the average value (in volts, accurate to two decimal places) of the voltage at node X, is _____.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q44. GATE ECE 2017 Set 2

Open solution

The state diagram of a finite state machine (FSM) designed to detect an overlapping sequence of three bits is shown in the figure. The FSM has an input 'In' and an output 'Out'. The initial state of the FSM is S_0.

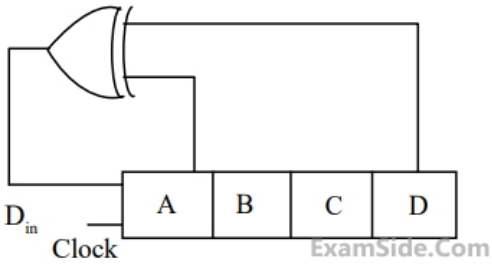


If the input sequence is 10101101001101, starting with the left-most bit, then the number of times 'Out' will be 1 is _____.

Q45. GATE ECE 2017 Set 1

Open solution

A 4-bit shift register circuit configured for right-shift operation is $D_{in} \rightarrow A$, $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$, is shown. If the present state of the shift register is ABCD = 1101, the number of clock cycles required to reach the state ABCD = 1111 is

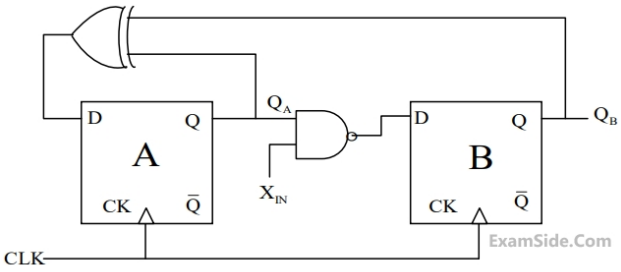


Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q46. GATE ECE 2017 Set 1

Open solution

A finite state machine (FSM) is implemented using the D flip-flops A and B and logic gates, as shown in the figure below. The four possible states of the FSM are $Q_A\ Q_B$ = 00, 01, 10, and 11.



Assume that X_{IN} is held at a constant logic level throughout the operation of the FSM. When the FSM is initialized to the state $Q_A\ Q_B$ = 00 and clocked, after a few clock cycles, it starts cycling through

(A) all of the four possible states if X_{IN} = 1

(B) three of the four possible states if X_{IN} = 0

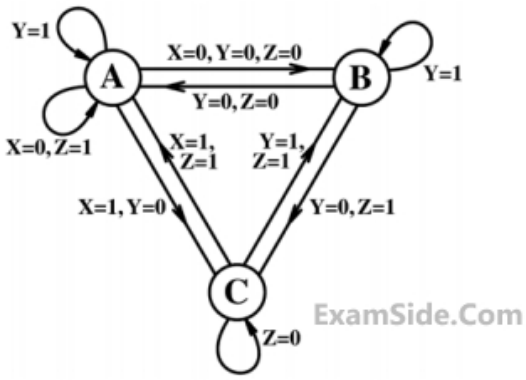
(C) only two of the four possible states if X_{IN} =1

(D) only two of the four possible states if X_{IN} = 0

Q47. GATE ECE 2016 Set 2

Open solution

The state transition diagram for a finite state machine with states A, B and C, and binary inputs X, Y and Z, is shown in the figure.



Which one of the following statements is correct?

(A) Transitions from State A are ambiguously defined.

(B) Transitions from State B are ambiguously defined.

(C) Transitions from State C are ambiguously defined.

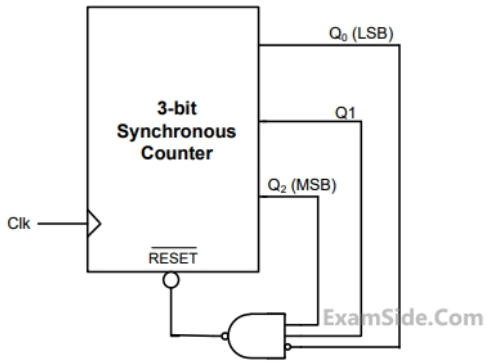
(D) All of the state transitions are defined unambiguously.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q48. GATE ECE 2016 Set 2

Open solution

For the circuit shown in the figure, the delay of the bubbled NAND gate is 2ns and that of the counter is assumed to be zero



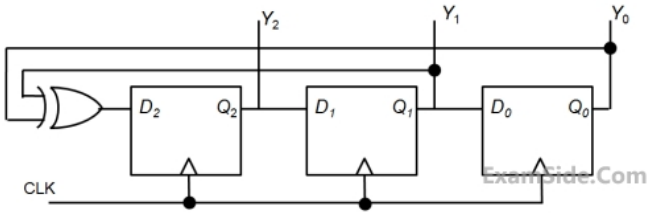
If the clock (Clk) frequency is 1 GHz, then the counter behaves as a

(A) mod-5 counter (B) mod-6 counter (C) mod-7 counter (D) mod-8 counter

Q49. GATE ECE 2015 Set 3

Open solution

A three bit pseudo random number generator is shown. Initially the value of output Y = Y_2 Y_1 Y_0 is set to 111. The value of output Y after three clock cycles is

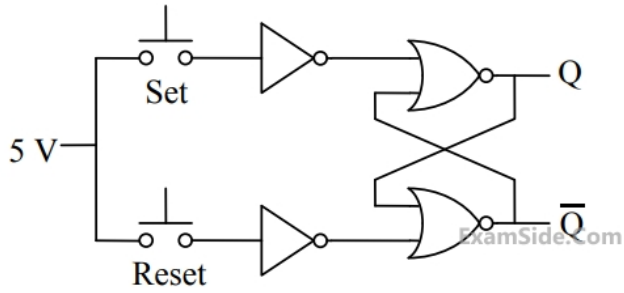


(A) 000 (B) 001 (C) 010 (D) 100

Q50. GATE ECE 2015 Set 3

Open solution

An SR latch is implemented using TTL gates as shown in the figure. The set and reset pulse inputs are provided using the push-button switches. It is observed that the circuit fails to work as desired. The SR latch can be made functional by changing



(A) NOR gates to NAND gates

(B) inverts to buffers

(C) NOR gates to NAND gates and inverters to buffers

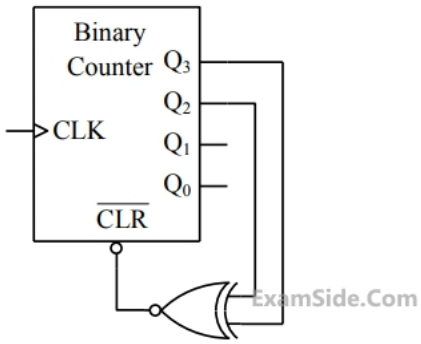
(D) 5 V to ground

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q51. GATE ECE 2015 Set 2

The figure shows a binary counter with synchronous clear input. With the decoding logic shown, the counter works as a

Open solution

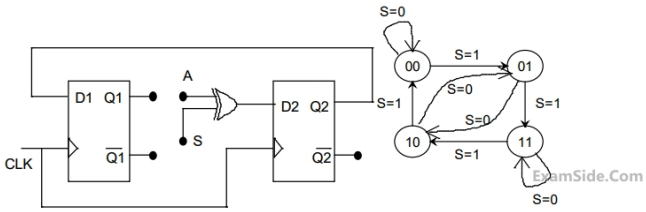


(A) mod - 2 counter (B) mod - 4 counter (C) mod - 5 counter (D) mod - 6 counter

Q52. GATE ECE 2014 Set 1

The digital logic shown in the figure satisfies the given state diagram when Q_1 is connected to input A of the XOR gate.

Open solution



Suppose the XOR gate is replaced by an XNOR gate. Which one of the following options preserves the state diagram?

(A) Input A is connected to $\overline{Q}_2$

(B) Input A is connected to Q_2

(C) Input A is connected to $\overline{Q}_1$ and S is complemented

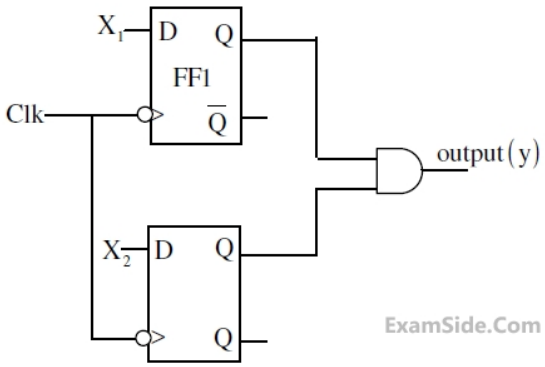
(D) Input A is connected to $\overline{Q}_1$

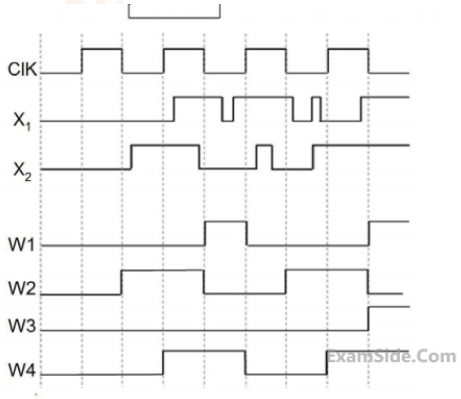
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q53. GATE ECE 2014 Set 2

Open solution

In the circuit shown, choose the correct timing diagram of the output (y) from the given waveforms W_1, W_2, W_3 and W_4.



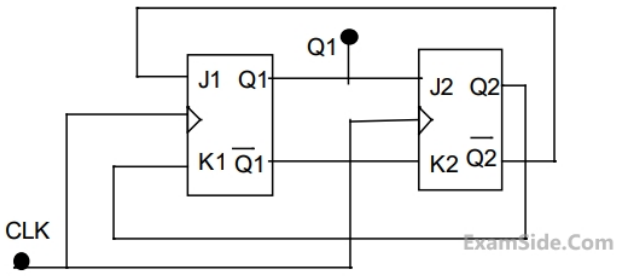


(A) W_1 (B) W_2 (C) W_3 (D) W_4

Q54. GATE ECE 2014 Set 2

Open solution

The outputs of the two flip-flops Q_1, Q_2 in the figure shown are initialized to 0, 0. The sequence generated at Q_1 upon application of clock signal is



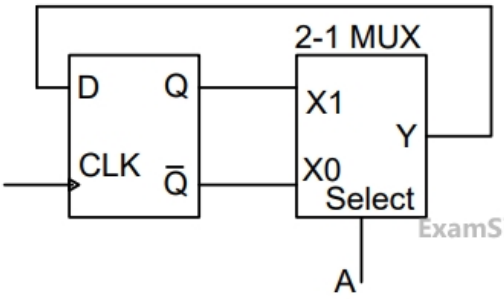
(A) 01110 (B) 01010 (C) 00110 (D) 01100

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

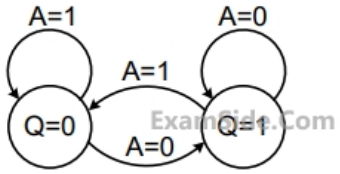
Q55. GATE ECE 2012

Open solution

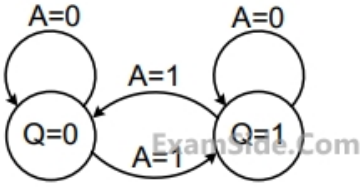
The state transition diagram for the logic circuit shown is



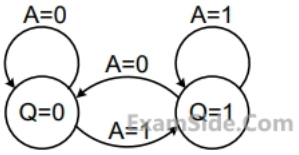
(A)



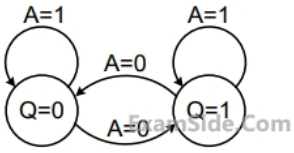
(B)



(C)



(D)



Q56. GATE ECE 2011

Open solution

Two D flip-flops are connected as a synchronous counter that goes through the following Q_B Q_A sequence $00 \rightarrow 11 \rightarrow 01 \rightarrow 10 \rightarrow 00 \rightarrow \ldots\ldots$

(A) $D_A = Q_{B,}\, D_B = Q_A$

(B) $D_A = \overline{Q}_A,\, D_B = \overline{Q}_B$

(C) $D_A = \left(Q_A \overline{Q}_B + \overline{Q}_A Q_B\right),\; D_B = Q_A$

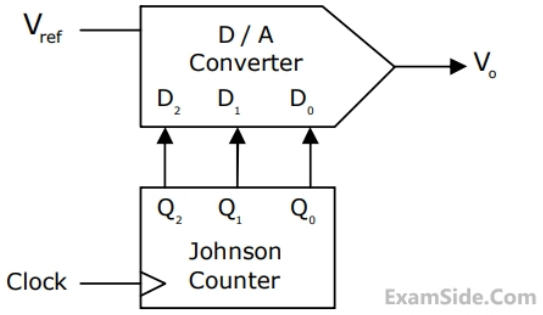
(D) $D_A = \left(Q_A Q_B + \overline{Q}_A \overline{Q}_B\right),\; D_B = \overline{Q}_B$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

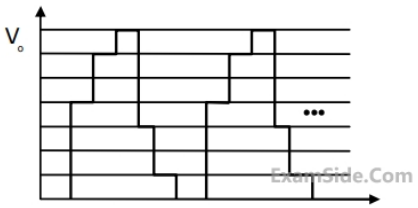
Q57. GATE ECE 2011

Open solution

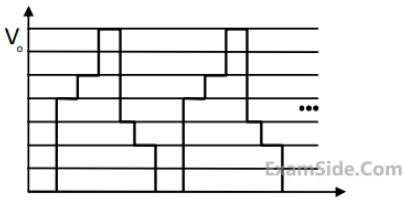
The output of a 3-stage Johnson (twisted-ring) counter is fed to a digital-to-analog (D/A) converter as shown in the figure below. Assume all the states of the counter to be unset initially. The waveform which represents the D/A converter output V_o is



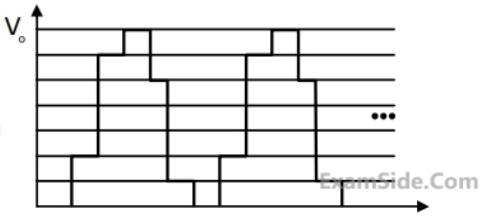
(A)



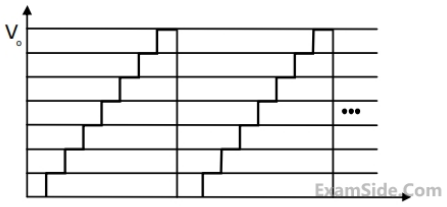
(B)



(C)



(D)

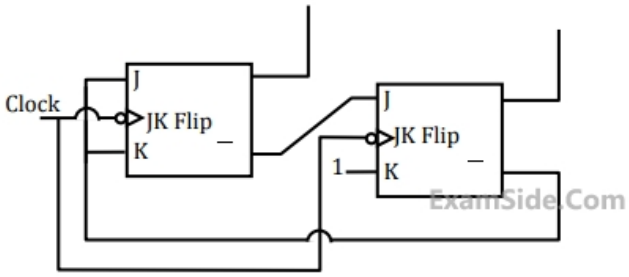


Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q58. GATE ECE 2009

What are the counting states (Q_1, Q_2) for the counter shown in the figure below?

Open solution



(A) 11, 10, 00, 11, 10,.....

(B) 01,1011,00,01.....,

(C) 00,11,01,10,00.....

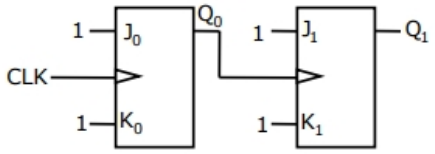
(D) 01,10,00,01,10.....

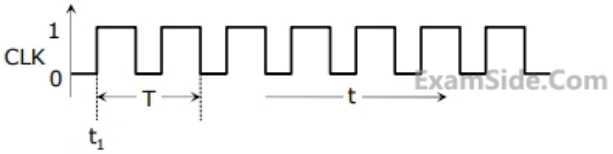
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q59. GATE ECE 2008

Open solution

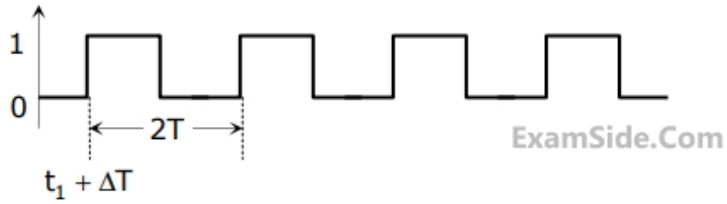
For each of the positive edge-triggered J-K flip flop used in the following figure, the propagation delay is ΔT



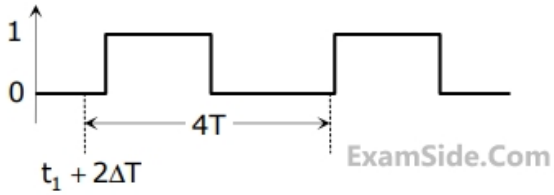


Which of the following waveforms correctly represents the output at Q_1?

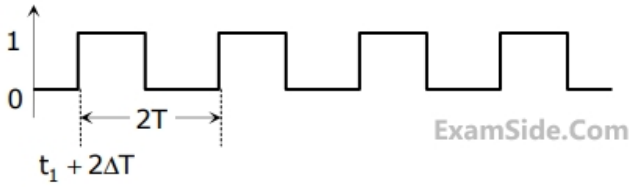
(A)



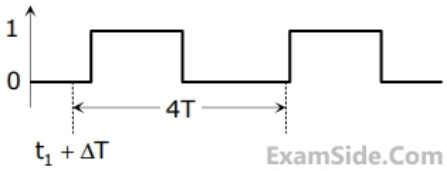
(B)



(C)



(D)

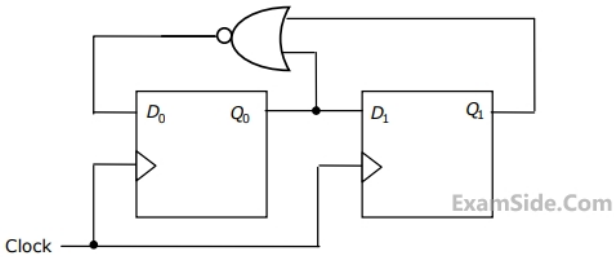


Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q60. GATE ECE 2007

Open solution

For the circuit shown, the counter state (Q_1 Q_0) follows the sequence



(A) 00,01,10,11,00.....

(B) 00,01,10,00,01.....

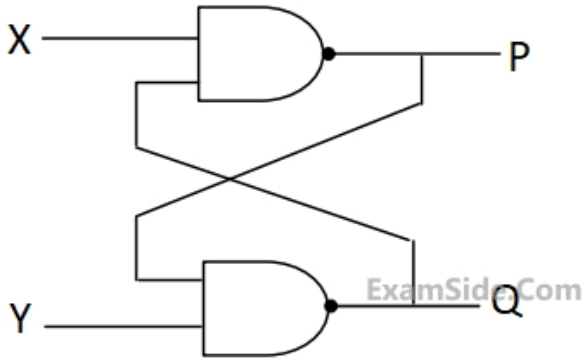
(C) 00,01,11,00,01.....

(D) 00,10,11,00,10.....

Q61. GATE ECE 2007

Open solution

The following binary values were applied to the X and Y inputs of the NAND latch shown in the figure in the sequence indicated below: X=0, Y=1; X=0, Y=0; X=1, Y=1. The corresponding stable P, Q outputs will be



(A) P = 1, Q = 0; P = 1, Q = 0; P = 1, Q = 0 or P = 0, Q = 1

(B) P = 1, Q = 0; P = 0, Q = 1; or P = 0, Q = 1; P = 0, Q = 1

(C) P = 1, Q = 0; P = 1, Q = 1; P = 1, Q = 0 or P = 0, Q = 1

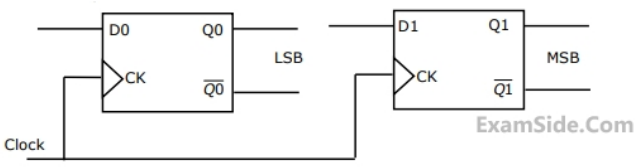
(D) P = 1, Q = 0; P = 1, Q = 1; P = 1, Q = 1

Q62. GATE ECE 2006

Open solution

Two D-flip-flops, as shown below, are to be connected as a synchronous counter that goes through the following Q_1Q_0 sequence $00 \rightarrow 01 \rightarrow 11 \rightarrow 10 \rightarrow 00 \rightarrow \ldots\ldots$

The inputs D_0 and D_1 respectively should be connected as



(A) $\overline{Q_1}$ *and* Q_0

(B) $\overline{Q_0}$ *and* Q_1

(C) $\overline{Q_1}Q_0$ *and* $\overline{Q_1}Q_0$

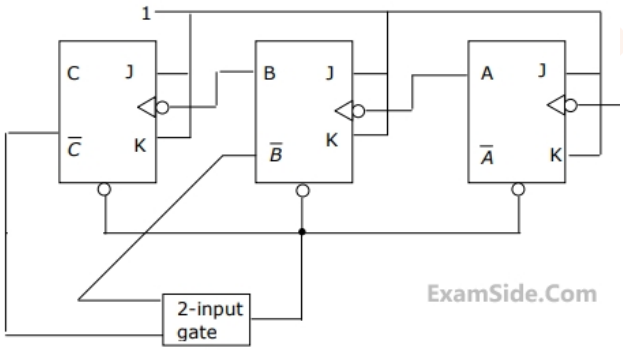
(D) $\overline{Q_1Q_0}$ *and* Q_1Q_0

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q63. GATE ECE 2004

In the modulo-6 ripple counter shown in the figure, the output of the 2-input gate is used to clear the J-K flip-flops. The 2-input gate is

Open solution



(A) a NAND gate (B) a NOR gate (C) an OR gate (D) an AND gate

Q64. GATE ECE 2003

A 4 bit ripple counter and a 4 bit synchronous counter are made using flip-flops having a propagation delay of 10 ns each. If the worst case delay in the ripple counter and the synchronous counter be R and S respectively, then

Open solution

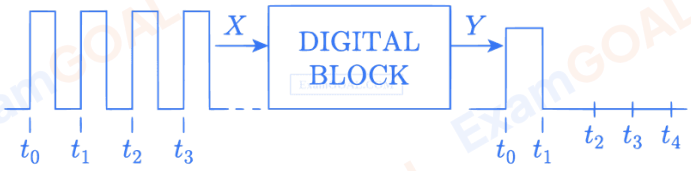
(A) R = 10ns, S = 40ns
(B) R = 40ns, S = 10ns
(C) R = 10ns, S = 30ns
(D) R = 30ns, S = 10ns

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q65. GATE ECE 2001

Open solution

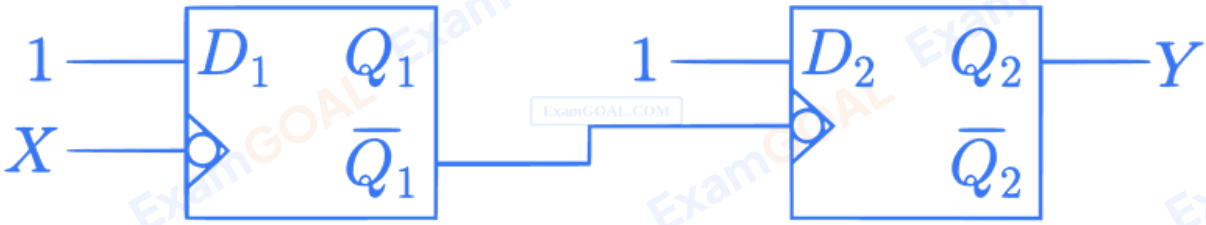
The digital block in the figure is realized using two positive edge triggered D flip-flops. Assume that for $t < t_0$, $Q_1 = Q_2 = 0$. The circuit in the digital block is given by



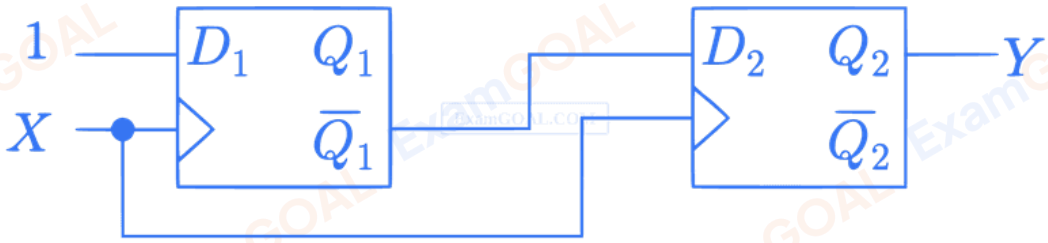
(A)



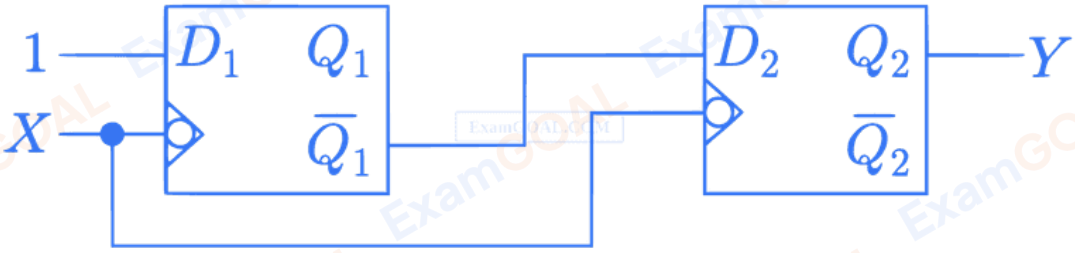
(B)



(C)



(D)

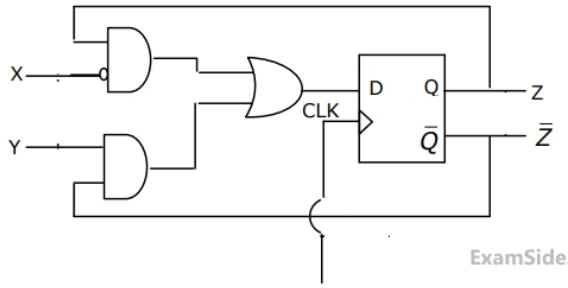


Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q66. GATE ECE 2000

A sequential circuit using D flip-flop and logic gates is shown in the figure, where X and Y are the inputs and Z is the output. The circuit is

Open solution

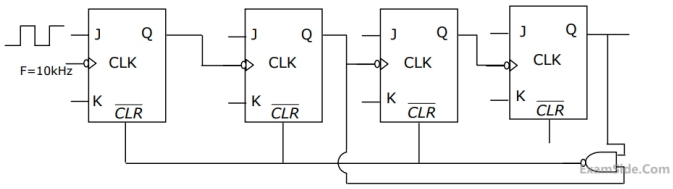


(A) S–R Flip-Flop with inputs X = R and Y = S
(B) S–R Flip-Flop with inputs X = S and Y = R
(C) J–K Flip-Flop with inputs X = J and Y = K
(D) J–K Flip-Flop with inputs X = K and Y = J

Q67. GATE ECE 2000

In the figure, the J and K inputs of all the four Flip-Flops are made high. The frequency of the signal at output Y is

Open solution

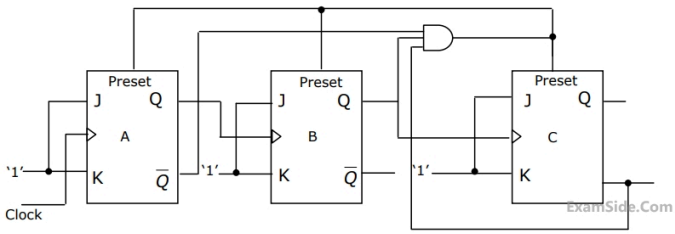


(A) 0.833 KHz (B) 1.0 KHz (C) 0.91 KHz (D) 0.77 KHZ

Q68. GATE ECE 1999

The ripple counter shown in the figure works as a

Open solution



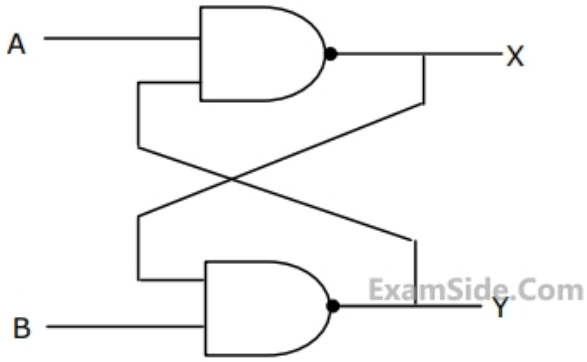
(A) mod – 3 up counter
(B) mod – 5 up counter
(C) mod – 3 down counter
(D) mod – 5 down counter

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q69. GATE ECE 1998

Open solution

In figure, A = 1 and B = 1. The input B is now replaced by a sequence 101010 ___, the outputs x and y will be



(A) fixed at 0 and 1, respectively

(B) x = 1010 ------while y = 0101 ------

(C) x = 1010 ----and y = 1010 ----

(D) fixed at 1 and 0, respectively

Marks 5 6 questions

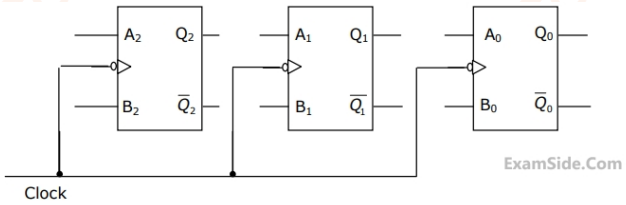
Q70. GATE ECE 2002

Open solution

It is required to design a binary mod-5 synchronus counter using AB flip-flops such that the output $Q_2Q_1Q_0$ changes as $000 \rightarrow 001 \rightarrow 010$ .....and so on. The excitation table for the AB flip-flops is given in the table

A	B	Q_{n+1}
0	0	1
0	1	$\overline{Q}_n$
1	0	Q_n
1	1	0

ExamSide.Com



(a) Write down the state table for the mod-5 counter.

(b)Obtain simplified SOP expressions for the inputs A_2, B_2, A_1, B_1, A_0 and B_0 in terms of Q_2, Q_1, Q and their complements.

(c) Hence, complete the circuit diagram for the mod-5 counter given in the figure using minimum number of 2-input NAND-gate only.

Q71. GATE ECE 1999

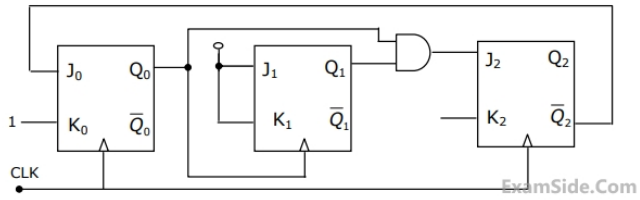
Open solution

The circuit diagram of a synchronous counter is shown in the figure. Determine the sequence of states of the counter assuming that the initial state is '000'. Give your answer in a tabular form showing the present state $Q_{A(n)}$, $Q_{B(n)}$, $Q_{C(n)}$, J-K inputs (J_A, K_A, J_B, K_B, J_C, K_C) and the next state $Q_{A(n+1)}$, $Q_{B(n+1)}$, $Q_{C(n+1)}$ From the table, determine the modulus of the counter.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q72. GATE ECE 1998

The mod-5 counter shown in figure counts through states $Q_2\ Q_1\ Q_0$ = 000, 001, 010, 011 and 100.



(a) Will the counter lockout if it happens to be in any one of the unused states?

(b) Find the maximum rate at which the counter will operate satisfactorily. Assume the propagation delays of flip-flop and AND gate to be t_F and t_A

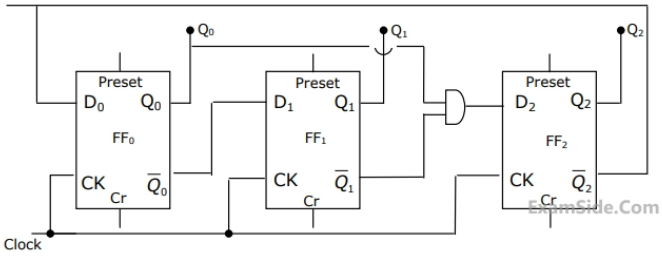
Open solution

Q73. GATE ECE 1997

A sequence generator is shown in figure. The counter status ($Q_0\ Q_1\ Q_3$) is intialized to 010 using preset/clear inputs. The Clock has a period of 50ns and transitions take place at the rising clock edge.

(a) Give the sequence generated at Q_0 till it repeats.

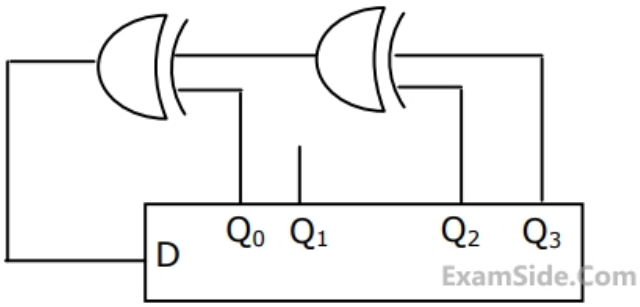
(b) What is the repetition rate for the generated sequence?



Open solution

Q74. GATE ECE 1996

A 4-bit shift register, which shifts 1 bit to the right at every clock pulse, is intialized to values (1000) for ($Q_0Q_1Q_2Q_3$). The D input is derived from Q_0, Q_2 and Q_3 through two XOR gates as shown in figure.



(a) Write the 4-bit values ($Q_0Q_1Q_2Q_3$) after each clock pulse till the pattern (1000) reappears on ($Q_0Q_1Q_2Q_3$).

(b) To what values should the shift register be intialized so that the pattern (1001) occurs after the first clock pulse?

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q75. GATE ECE 1996

Open solution

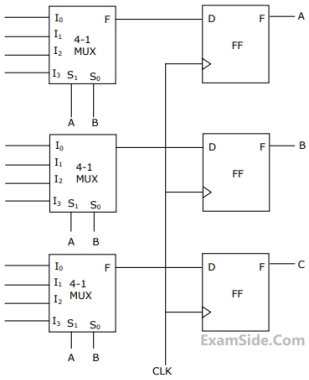
A state machine is required to cycle through the following sequence of states:

ABC

→000→010→111→100→011→101

ExamSide.Com

One possible implementation of the state machine is shown figure. Specify what signals should be applied to each of the multiplexer inputs



Marks 8 4 questions

Q76. GATE ECE 1992

Open solution

A new clocked "X-Y" flip-flop is defined with two inputs, X and Y in addition to the clock input. The flip-flop functions as follows:

If XY=00, the flip-flop changes state with each clock pulse.

If XY=01, the flip-flop state Q becomes 1 with the next clock pulse.

If XY=10, the flip-flop state Q becomes 0 with the next clock pulse.

If XY=11, no change of state occurs with the clock pulse.

(a) Write the Truth table for the X-Y flip flop

(b) Write the Excitation table for the X-Y flip flop

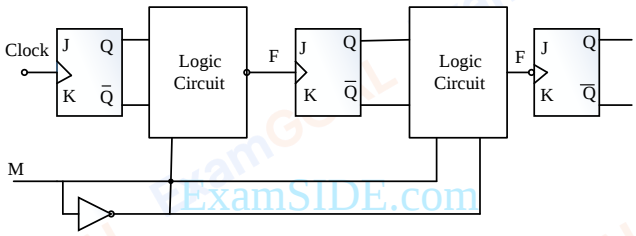
(c) It is desired to convert a J-K flip flop into the X-Y flip flop by adding some external gates, if necessary. Draw a circuit to show how you will implement the X-Y flip-flop using a J-K flip-flop.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q77. GATE ECE 1988

The circuit shown below uses TTL flip-flops. The flip-flops are triggered at the negative transitions of the clock. It is desired that when M = 1 the circuit should function as an up-counter (in 8421 BCD) and when M=0, as a down-counter. Design the combinational circuit interposed between the flip-flops so that the circuit works as desired. (i.e. find F as a function of Q,$\overline{Q}$, M, $\overline{M}$).

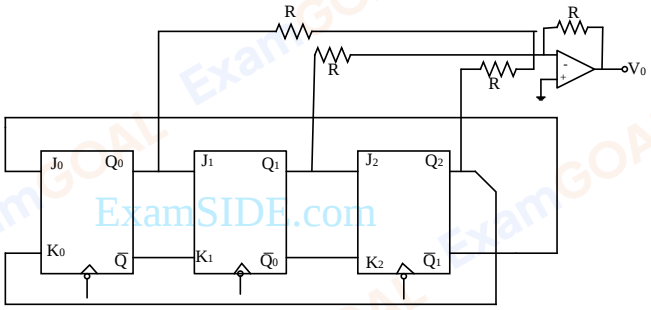
Open solution



Q78. GATE ECE 1988

For the circuit shown in the figure below, sketch V_0 against time. Assume that all flip-flops are reset to zero before the clock is applied.

Open solution



Q79. GATE ECE 1987

A 2-input up/down synchrconous counter using two toggle flip-flops is shown in Fig.1. The counter's sequence is to be controlled by the input M as follows:
For M=1, sequence of Q_1, Q_0 is ..00, 01, 10, 11, 00, 01.....
For M=0, sequence of Q_1, Q_0 is ..00, 11, 10, 01, 00, 11.....
(a)Design the necessary feedback logic for T_1 and T_0.

(b)Realize the feesback logic using inverters and 4-input multiplexers only. Use Q_1 and Q_0 as the control inputs of the multiplexer with Q_1 as the MSB.

Open solution

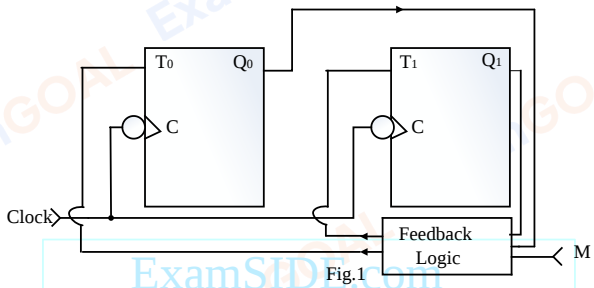


Fig.1

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Marks 1

1. **B**	2. **(5) (4.90 to 5.10)**	3. **A**	4. **5 to 5**	5. **30**
6. **B**	7. **1.5**	8. **A**	9. **7**	10. **62.5**
11. **D**	12. **D**	13. **A**	14. **A**	15. **D**
16. **C**	17. **B**	18. **C**	19. **B**	20. **D**
21. **C**	22. **D**	23. **D**	24. **C**	25. **faster**
26. **100**	27. **C**	28. **R, S**	29. **C**	30. **C**
31. **A**	32. **B**			

Marks 2

33. **A**	34. **A**	35. **200**	36. **B**	37. **250**
38. **A**	39. **A, C**	40. **4**	41. **77**	42. **D**
43. **0.82 to 0.86**	44. **4**	45. **10**	46. **D**	47. **C**
48. **D**	49. **D**	50. **D**	51. **C**	52. **D**
53. **C**	54. **D**	55. **D**	56. **D**	57. **A**
58. **A**	59. **B**	60. **B**	61. **C**	62. **A**
63. **C**	64. **B**	65. **C**	66. **D**	67. **A**
68. **D**	69. **A**			

Marks 5

70. **(b)** $A_2 = \overline{Q_2Q_0}$ $B_2 = 1$ $A_1 = \overline{Q_0}$ $B_1 = Q_0$ $A_0 = Q_2$ $B_0 = 1$	71. **It is mod 6 counter**	72. **(a) Counter is lockout free** **(b) Maximum clock rate =** $\frac{1}{2t_F + t_A}$	73. **(a) Sequence generated at Q_0 is** →0→1→1→1→0→ **(b) 4MHz**	74. **(b) Shift register should intialized to 1100**
75. **Solve it.**				

Marks 8

76. **Solve it.**	77. **Solve it.**	78. **Solve it.**	79. **Solve it.**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs

ExamGOAL
LEARNING APP

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.

GET IT ON Google Play